Charlie Chen

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EDUCATION

Harvard University Cambridge, MA

Bachelor of Science in Electrical Engineering, Minor in Computer Science | GPA: 3.97/4.0 May 2027

- **MIT Cross-Registration:** Solid-State Circuits, Semiconductor Electronic Circuits, Intro to Machine Learning
- **Relevant Courses:** Circuits Devices & Transduction, Signals & Communications, Computing Hardware, Systems & Control, Systems Programming, Data Structures & Algorithms

TECHNICAL SKILLS

EDA & Design Tools: Altium Designer, LTspice, Cadence Virtuoso, Spectre, KiCad

Lab & Test: Oscilloscope, Function Generator, Spectrum Analyzer, Power Supply, DMM, Soldering, Board Bring-Up

HDL & Digital: SystemVerilog, Xilinx Vivado, FPGA

Programming: Python, C, MATLAB, Git, Linux

EXPERIENCE

General Dynamics Applied Physical Sciences | Electrical Engineering Intern | Pawcatuck, CT May – Aug 2026

- Expanded underwater acoustic communication system to support 2 additional ~10kHz differential bands at ~5kV / 5A
- Routed amplifier outputs to frequency-appropriate magnetics & hydrophones via opto-isolated control signals
- Owned schematic capture, layout, library, & DRC across 3 PCB versions (4-layer, 2×4″) with varying switching topologies
- Collaborated with mechanical team to adjust layout & mounting configurations to meet spatial & thermal constraints
- Designed hydrophone-test board with an overcurrent latching-protection circuit utilizing comparator hysteresis

ECE1410: Computing Hardware | Teaching Fellow | Cambridge, MA Jan 2026 – Present

- Rewrite problem sets covering CMOS transistors, digital logic, FSMs, memory, RTL debugging & testbench writing
- Hold weekly office hours & review sessions, grade problem sets for 50 students, conduct FPGA hardware lab interviews

Harvard Ability Lab | Undergraduate Researcher | Cambridge, MA June – Nov 2025

- Prototyped wearable mounting & data recording system to collect 10,000+ frames for supernumerary-arm VLA study
- Coded custom firmware & camera handler to process & stream video into teleoperation, recording, & inference pipelines

ENGINEERING PROJECTS

Analog IC Tapeout — Differential Amplifier (TSMC 65nm) | Cadence Virtuoso, Spectre

- Designed transistor-level differential amp in TSMC 65nm meeting gain (>26 dB), power (<500µW), swing (>300mV) specs
- Completed layout with biasing network using hierarchical design flow; DRC/LVS-clean, sent to TSMC for fabrication
- Verified gain, bandwidth, & switching (Ron<100Ω, Roff>10MΩ) behavior via AC/DC/transient testbenches in Spectre

Reverb Karaoke Machine | Mixed-Signal Design, Analog Signal Conditioning, ADC/DAC

- Engineered analog signal-conditioning chain with input biasing, anti-alias filtering & output reconstruction filtering
- Implemented custom 10-bit R-2R DAC & SAR ADC, evaluating architecture as an alternative to on-chip MCU ADC
- Measured frequency response & end-to-end signal quality, diagnosing noise, clipping, & filtering issues during bring-up

NAND-Only Logic Gate Demo PCB | PCB Design & Bring-Up, Power & Signal Integrity, KiCad

- Designed, assembled, & tested 2-layer PCB implementing all 6 basic logic gates from NAND with user inputs
- Ensured power & signal integrity with per-IC decoupling & solid GND plane, suppressing VDD droop & GND bounce

MIPS Multicycle Processor & Assembler | SystemVerilog, RTL Design, FPGA, FSM Control, Testbenches

- Architected multicycle CPU to reuse shared datapath resources, trading control complexity for hardware efficiency
- Implemented FSM control path to sequence register, ALU, memory, & PC operations across instruction types
- Developed MIPS assembler & testbenches, verifying cycle-accurate CPU behavior via waveform analysis

ADDITIONAL SKILLS

Fluent Languages: English, Mandarin

Interests: Hiking, Cooking, Running, Speedcubing